

ASSIGNMENT No: 4

NAME : SUHANI GARG

PRN : 25070122194

BATCH: CSE B3

ACADEMIC YEAR :2025-29

TITLE: Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

CODE

```
public class WrapperClassSymbi
{
    public static void main(String[] args)
    {
        // Primitive to Wrapper (Autoboxing)

        int num = 100;
        Integer intObj = num;

        double value = 25.75;
        Double doubleObj = value;

        // Wrapper to Primitive (Unboxing)

        int primitiveInt = intObj;
        double primitiveDouble = doubleObj;

        // String to Wrapper

        Integer number = Integer.valueOf("200");
        Double decimal = Double.valueOf("45.5");

        // Wrapper to String

        String str1 = intObj.toString();
        String str2 = doubleObj.toString();

        // Basic Operations

        int sum = intObj + number;
        double product = doubleObj * decimal;

        System.out.println("Primitive int: " + primitiveInt);
        System.out.println("Primitive double: " + primitiveDouble);

        System.out.println("Integer Object: " + intObj);
        System.out.println("Double Object: " + doubleObj);

        System.out.println("Converted from String (Integer): " + number);
        System.out.println("Converted from String (Double): " + decimal);

        System.out.println("Integer to String: " + str1);
        System.out.println("Double to String: " + str2);

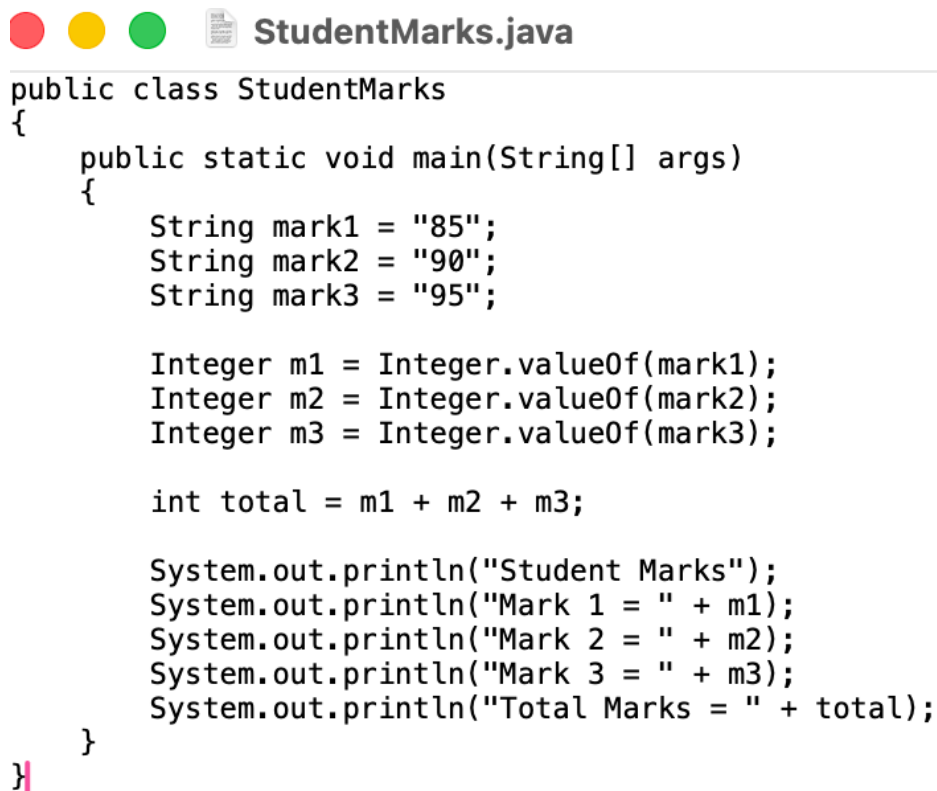
        System.out.println("Sum: " + sum);
        System.out.println("Product: " + product);
    }
}
```

OUTPUT

```
[suhanigarg670@Suhanis-MacBook-Air desktop % javac WrapperClassSymbi.java
[suhanigarg670@Suhanis-MacBook-Air desktop % java WrapperClassSymbi
Primitive int: 100
Primitive double: 25.75
Integer Object: 100
Double Object: 25.75
Converted from String (Integer): 200
Converted from String (Double): 45.5
Integer to String: 100
Double to String: 25.75
Sum: 300
Product: 1171.625
suhanigarg670@Suhanis-MacBook-Air desktop % █
```

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.



```
StudentMarks.java
public class StudentMarks
{
    public static void main(String[] args)
    {
        String mark1 = "85";
        String mark2 = "90";
        String mark3 = "95";

        Integer m1 = Integer.valueOf(mark1);
        Integer m2 = Integer.valueOf(mark2);
        Integer m3 = Integer.valueOf(mark3);

        int total = m1 + m2 + m3;

        System.out.println("Student Marks");
        System.out.println("Mark 1 = " + m1);
        System.out.println("Mark 2 = " + m2);
        System.out.println("Mark 3 = " + m3);
        System.out.println("Total Marks = " + total);
    }
}
```

OUTPUT

```
[suhanigarg670@Suhanis-MacBook-Air ~ % cd desktop
[suhanigarg670@Suhanis-MacBook-Air desktop % javac StudentMarks.java
[suhanigarg670@Suhanis-MacBook-Air desktop % java StudentMarks
Student Marks
Mark 1 = 85
Mark 2 = 90
Mark 3 = 95
Total Marks = 270
suhanigarg670@Suhanis-MacBook-Air desktop % █
```

2. Develop an Employee Payroll System that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

CODE

```
public class EmployeePayroll
{
    public static void main(String[] args)
    {
        String empId = "101";
        String salary = "50000";
        String bonus = "5000";

        Integer id = Integer.valueOf(empId);
        Double basicSalary = Double.valueOf(salary);
        Double bonusAmount = Double.valueOf(bonus);

        if (basicSalary > 0)
        {
            double netSalary = basicSalary + bonusAmount;

            System.out.println("Employee ID = " + id);
            System.out.println("Basic Salary = " + basicSalary);
            System.out.println("Bonus = " + bonusAmount);
            System.out.println("Net Salary = " + netSalary);
        }
        else
        {
            System.out.println("Invalid Salary");
        }
    }
}
```

OUTPUT

```
[suhanigarg670@Suhanis-MacBook-Air ~ % cd desktop
[suhanigarg670@Suhanis-MacBook-Air desktop % javac EmployeePayroll.java
[suhanigarg670@Suhanis-MacBook-Air desktop % java EmployeePayroll
Employee ID = 101
Basic Salary = 50000.0
Bonus = 5000.0
Net Salary = 55000.0
suhanigarg670@Suhanis-MacBook-Air desktop % █
```

CONCLUSION

This assignment demonstrates the use of wrapper classes in Java for data conversion between String, primitive data types, and wrapper objects. It also shows how wrapper classes are used to perform calculations and basic operations, helping in better understanding of Java programming concepts.

